

Praxia Update

Daily orthopaedic literature digest · Saturday, 13 June 2026 · 49 new articles across 10 journals



American Journal of Sports Medicine

June 2026 · online ahead of print · 1 new article

Partial Tenotomy and Reattachment for Chronic Proximal Hamstring Tendinopathy: Functional Outcomes, Return to Sport, and Patient Satisfaction.

Lefèvre N, Moussa MK, Valentin E, Hadjam T, Gerometta A, Grimaud O, Meyer A, Bohu Y et al.

BACKGROUND: Chronic proximal hamstring tendinopathy (PHT) can cause significant pain and functional impairment, particularly in active individuals. While nonoperative management is the first-line treatment, some patients with persistent symptoms may require surgical intervention.

PURPOSE: To evaluate the functional outcomes, return-to-sport rates, and complication profiles of patients undergoing partial tenotomy and reinsertion for chronic PHT.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This retrospective cohort study targeted patients who underwent partial tenotomy and reattachment of the proximal hamstring tendons between April 2010 and September 2022. The indication for surgery was refractory chronic PHT symptoms for >6 months. The primary outcome measure was the Parisian Hamstring Avulsion Score (PHAS). Secondary outcomes included the Tegner Activity Scale score, University of California-Los Angeles (UCLA) Activity Scale score, return to sport, postoperative patient satisfaction, and the rate of complications. In this study, the terms "partial involvement" or "complete involvement" refer to the extent of magnetic resonance imaging signal changes across the tendon, rather than to true tendon ruptures or avulsions.

RESULTS: The study included 42 patients with a mean age of 50.4 years (SD, 10.4 years). The mean follow-up was 4.1 years (range, 2.0-12.6 years [...])

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Long-term Oncologic Outcomes After Unplanned Soft Tissue Sarcoma Surgery: Implications for Surveillance Strategies.

Jahn J, Dean KK, Travis LM, Daftari M, Costello JP, Shloush M, Brutti J, Shae J et al.

BACKGROUND: Unplanned excision refers to the resection of soft tissue sarcomas without appropriate preoperative imaging, histologic confirmation, or oncologic planning. Although the short-term oncologic risks of unplanned excisions are well described, data on long-term outcomes remain limited.

QUESTIONS/PURPOSES: After controlling for tumor grade, clinical stage, Charlson comorbidity index (CCI), and surgical margin status, is unplanned excision independently associated with (1) worse overall survival, (2) local recurrence-free survival, and (3) metastatic disease-free survival compared with planned excision?

METHODS: In this study of 1100 patient records initially identified by Current Procedural Terminology code query, 403 were identified as duplicate records, 253 were excluded for insufficient baseline data or lack of cancer center entry, and 48 were excluded for insufficient follow-up or not undergoing surgery at the institution, yielding 396 patients, 279 who underwent planned excision and 117 who underwent unplanned excision. We defined an unplanned excision as resection of a soft tissue mass performed without appropriate preoperative advanced imaging (such as MRI) or histologic tissue diagnosis (such as core needle biopsy), or when imaging and biopsy findings were misinterpreted as benign, leading to resection without oncologic planning. The mean $\pm$ SD age was similar b [...]

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Letter to the Editor: Editorial: AAOS Orthobiologics Registry-Sometimes, More is Less.

Wang X, Shen W

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Does Liposomal Bupivacaine Provide Superior Postoperative Analgesia Compared With Ropivacaine With Dexamethasone in Suprainguinal Fascia Iliaca Compartment Block for THA? A Randomized Controlled Trial.

Wu W, Hu X, Yang C, Sun Y, Li J, Wang L

BACKGROUND: Choosing the most effective pain management is important for enhancing functional recovery after surgery. The fascia iliaca compartment block (FICB) is recommended for analgesia after THA, but whether liposomal bupivacaine is more effective for this application remains uncertain.

QUESTIONS/PURPOSES: (1) Did a single-injection suprainguinal FICB using liposomal bupivacaine provide superior pain control by a clinically important margin? (2) What was the effect of liposomal bupivacaine on perioperative opioid consumption? (3) How did liposomal bupivacaine affect the duration of the sensory block and quadriceps motor weakness in the operated limb?

METHODS: This was a prospective, parallel-group, assessor-blinded, single-center, randomized controlled clinical trial. Patients who underwent unilateral primary THA via the posterolateral approach were included in the study. Exclusion criteria included a BMI of > 32.5 kg/m² (which is the diagnostic criterion for moderate obesity in China), contraindications to peripheral nerve block, known allergy to study medications, psychiatric or cognitive disorders that could interfere with pain assessment, severe hepatic or renal dysfunction, and chronic pain necessitating long-term opioid therapy. According to these criteria, 76% (60 of 79) of screened patients met eligibility requirements and were subsequently randomized into either [...]

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Is Neighborhood Socioeconomic Deprivation Associated With Outcomes Following Primary THA and TKA? A Systematic Review Utilizing the Area Deprivation Index.

Furyes AR, Elmenawi KA, Hackett L, Deren ME, Piuze NS

BACKGROUND: Socioeconomic disadvantage is increasingly recognized as an important factor associated with outcomes after total joint arthroplasty (TJA). The Area Deprivation Index (ADI) has emerged as a validated measure of socioeconomic disadvantage that specifically captures individual neighborhood levels of structural and material deprivation based on income, education, employment, and housing characteristics. Prior research has demonstrated ADI to be associated with surgical outcomes; however, its relationship to postoperative outcomes after THA and TKA, procedures in which outcomes depend highly on institutional and structural support, home environment, and accessible postacute care resources, has not been well synthesized. Prior studies provide a broad overview of heterogeneous measures of socioeconomic disadvantage that are not directly comparable.

QUESTIONS/PURPOSES: (1) Is a higher ADI associated with increased postoperative healthcare utilization after primary THA and TKA? (2) Is a higher ADI associated with differences in patient-reported outcome measures (PROMs)? (3) Is a higher ADI associated with increased postoperative complications? (4) Is ADI associated with differences in implant survivorship after TJA?

METHODS: A systematic review of four databases-Ovid Medline, Embase, Cochrane Library, and Web of Science Core Collection-was performed from their inception to [...]

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Editor's Spotlight/Take 5: How Does Traumatic Hand Injury Impact Patients in a Safety-net Healthcare System? A Mixed Methods Study.

Wongworawat MD

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Letter to the Editor: Guest Editorial: Recalling a Recall.

Doyle TR, Hennessy O, Keogh CJ, Hughes AJ, Kenny PJ, Sheridan GA, Murphy CG

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JBJS (American)

June 2026 · online ahead of print · 3 new articles

A System in Motion: The Evolution of Orthopaedic Care in Romania.

Dragosloveanu S, Nedelea DG, Scheau C

> Romanian orthopaedic practice has demonstrated a growing uptake of digital infrastructure, including the implementation of robotic-assisted surgical workflows and emerging applications of artificial intelligence.> Romania ranks among the European countries with the highest hospital bed capacity per capita, supporting the provision of sustained care for elderly patients, including those with fragility fractures and oncologic orthopaedic conditions and those undergoing complex revision surgeries.> The Romanian Arthroplasty Register was modernized and relaunched in 2025 as an initiative to align with other long-standing national European counterparts.

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Comparative Analysis of Immune Cell-Type Abundances in Periprosthetic Tissues Across Arthroplasty Failure Etiologies: Use of Transcriptomic Deconvolution.

Li Y, Wang F, Li Y, Mu W, Ji B, Zhang X, Cao L

BACKGROUND: Although cellularity is traditionally evaluated morphologically, an emerging transcriptome-sequencing-based algorithm enables simultaneous inference of cellular information. We studied whether cellularity profiles predicted using CIBERSORTx would (1) depict immune cell-type abundances in periprosthetic tissues across arthroplasty failure etiologies, and (2) provide prognostic value for identifying cases of periprosthetic joint infection (PJI).

METHODS: CIBERSORTx-derived cellularity profiles were evaluated in 185 periprosthetic tissue samples, including 135 from patients with PJI (64 males; median age, 66 years) and 50 from those with aseptic failure (AF) (36 males; median age, 62.5 years), that had been subjected to bulk RNA sequencing. Kaplan-Meier survival analysis was performed to assess prognostic outcomes in PJI.

RESULTS: Of the 22 evaluated cell types, 5 were significantly elevated in PJI cases: plasma cells, resting memory CD4+ T cells, CD8+ T cells, activated mast cells, and M1 macrophages (all $p < 0.05$ after Benjamini-Hochberg [BH] correction). Conversely, 3 cell types were significantly elevated in AF cases: gamma delta T cells, M0 macrophages, and M2 macrophages (all $p < 0.05$ after BH correction). Of the combined immune cell populations, total B cells, total T cells, and natural killer cells were significantly elevated in PJI cases, while total macroph [...]

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Minimal Movement Restrictions Do Not Increase Hip Dislocations Following Total Hip Arthroplasty: A Before-and-After Study of 10,357 Patients.

Klaassen AD, Willigenburg NW, Musters JWQ, Jager J, Poolman RW, Santeon Hip Osteoarthritis Group

BACKGROUND: Hip dislocation is the most common reason for early revision after total hip arthroplasty (THA). To prevent dislocation, patients are often prescribed movement restrictions following THA; however, these restrictions can be uncomfortable and limiting for patients and may delay a return to daily activities. The present study aimed to investigate whether a protocol with reduced movement restrictions following THA would lead to an increased incidence of early hip dislocations.

METHODS: In this uncontrolled, multicenter, before-and-after study, 2 groups were retrospectively compared. Patients who underwent a THA between January 1, 2015, and March 31, 2020, were included. Patients in the first group ($n = 7,666$) were prescribed strict movement restrictions following THA, and patients in the second group ($n = 2,691$) were prescribed minimal movement restrictions. Patient and prosthesis characteristics were collected, as well as data regarding hip dislocations within 90 days postoperatively as the primary outcome.

RESULTS: The incidence of early hip dislocations was not significantly different between movement groups, with 112 (1.46%) of 7,661 hips in the restricted group and 52 (1.93%) of 2,691 hips in the minimally restricted group experiencing a dislocation ($p = 0.093$).

CONCLUSIONS: The use of a protocol with minimal movement restrictions following THA did not increase [...]

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Propensity Score-Matched Comparison of Patient-Reported Outcomes Between Crowe I Developmental Dysplasia of the Hip and Primary Osteoarthritis Following Total Hip Arthroplasty.

Saluja A, Eschert J, Ren R, Wong Z, Su E

BACKGROUND: Total hip arthroplasty (THA) is commonly performed for Crowe I developmental dysplasia of the hip (DDH) and primary osteoarthritis (OA), but comparative survivorship and patient-reported outcome measure (PROM) data remain limited with mixed findings. This study aimed to compare PROMs, revision rates, and implant survivorship between propensity-matched THA patients who had Crowe I DDH and OA.

METHODS: A retrospective review of a single surgeon's case log was performed to identify patients who had a diagnosis of DDH or OA who underwent THA. Propensity-score matching was utilized, yielding 105 DDH hips and 105 OA hips. Primary outcomes were modified Harris Hip Score (mHHS), Hip disability and Osteoarthritis Outcome Score - Joint Replacement (HOOS-JR), Visual Analog Scale (VAS) for pain, and University of California, Los Angeles Activity Scale (UCLA) collected preoperatively, at one year, and at minimum 2-year final follow-up. The secondary outcomes included implant survivorship via Kaplan-Meier analysis, multivariable logistic regression for minimal clinically important difference (MCID) and substantial clinical benefit (SCB) achievement, revision rates, and radiographic assessment of leg length discrepancy and acetabular inclination.

RESULTS: There were no significant between-group differences in any PROM at any time point (all $P > 0.05$). Analysis of covariance (ANCOVA) [...]

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What Factors Increase the Risk of Dislocation after Primary Total Hip Arthroplasty Using the Direct Anterior Approach?

Murphy MP, Forti CR, Ho H, Hopper RH, Hamilton WG

BACKGROUND: While the direct anterior approach (DAA) for total hip arthroplasty (THA) is associated with a lower risk of dislocation, this complication has not been eliminated and can be acutely disabling when it occurs. This study aimed to identify patient, surgical, and radiographic factors that contribute to dislocation risk for the DAA based on primary THAs performed over a 15-year period.

METHODS: Using an institutional database, we identified 8,658 DAA THAs performed by four surgeons from March 2009 through May 2024. A Cox proportional hazards regression was performed to determine how patient demographics and implant characteristics were associated with dislocation. Patients who dislocated were also compared to a matched control cohort without dislocation to evaluate the influence of radiographic parameters, including acetabular cup orientation, offset, limb length changes, and preoperative acetabular coverage.

RESULTS: The dislocation rate among all primary THAs was 0.5% (45 of 8,656). Women had a higher dislocation rate than men (0.7 versus 0.2%), and Cox regression identified an increased risk among women with 28 mm (hazard ratio (HR) 7.5; 95% confidence interval (CI) 3.2 to 17.6) and 32 mm heads (HR 3.8; 95% CI 1.9 to 7.5). Radiographic analysis showed no significant differences in total offset ($P = 0.13$) or limb length change ($P = 0.41$), but controls demonstrated s [...]

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Robotic Total Hip Arthroplasty in Atypical Hip Anatomy: Accuracy of Component Positioning and Clinical Outcomes in 192 Complex Cases.

Kayani B, Enson J, Donaldson J, Miles J, Newman S, Jayadev C, Stammers J, Skinner JA

BACKGROUND: The objectives of this study were to determine the accuracy of component positioning, patient satisfaction, functional outcomes, component survivorship, and complications of robotic total hip arthroplasty (THA) in patients who have atypical hip anatomy.

METHODS: This study included 192 robotic THAs performed in 182 patients for developmental dysplasia of the hip (n = 122), Leg-Calve-Perthes disease (n = 27), slipped capital femoral epiphysis (n = 20), previous acetabular fracture (n = 12), and skeletal dysplasia (n = 11). Predefined radiological outcomes, patient satisfaction, University of California at Los Angeles (UCLA) activity score, Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), Oxford Hip Score (OHS), Forgotten Joint Score (FJS), component survivorship, and any complications were recorded. The mean follow-up time was 3.8 ± 1.7 years (range, 2.1 to 5.4).

RESULTS: Robotic THA was associated with high levels of accuracy in executing the planned horizontal (root mean square error (RMSE) 1.5 ± 1.4 mm) and vertical centers of rotation (RMSE: 1.8 ± 1.7 mm), combined offset (RMSE: 2.9 ± 3.8 mm), and leg-length correction (RMSE: 1.5 ± 1.4 mm). Acetabular component positioning within Lewinnek's safe zones was 94.7%, and Callanan's safe zones was 93.8%. The median patient satisfaction score was 90 (interquartile range (IQR), 85 to 95), the med [...]

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Social Determinants of Health Z-Codes are Associated with Low Compliance with Patient-Reported Outcome Measures after Total Hip Arthroplasty.

Parry M, Ozdag Y, Kloc A, Salvato J, Mercuri JJ

INTRODUCTION: There is limited literature on compliance rates for completing patient-reported outcome measures (PROMs) after elective total hip arthroplasty (THA). Social Determinants of Health (SDOH) refers to social factors that are not classifiable elsewhere that can have major impact on healthcare outcomes. The purpose of this investigation was to assess the relationship between SDOH codes, demographic and surgical factors, and compliance rates for filling out PROMs after THA.

METHODS: This was a retrospective cohort study of elective THA cases performed between 2017 and 2022 at a single institution. Patients less than 18 years of age and patients who had less than two years of follow-up were excluded. The SDOH codes ("Z" codes), mental health codes ("F" codes), and compliance rates were recorded. The study population was binarized into two cohorts: any compliance (greater than 0%) versus no compliance (0%). Bivariate and multivariate regression (MRA) analyses were performed to assess compliance with PROMs against SDOH, demographic, and surgical factors. A total of 3,392 cases were included, of which 1,779 (52.5%) had completed at least one set of PROM questionnaires while 1,613 (47.5%) were non-compliant. Overall survey completion in the study period was 25.8%.

RESULTS: Results of the MRA indicated that having any Z-codes (odds ratio (OR) 0.355 [95% CI (confidence interv [...]

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Differences Between Inpatient and Outpatient Medicare Total Knee Arthroplasty: Substantial Clinical Benefit Thresholds on Knee Injury and Osteoarthritis Outcome Score for Joint Replacement.

Patel SV, Pasqualini I, Khan ST, Elmenawi KA, Zhang C, Piuizzi NS, Cleveland Clinic Adult Reconstruction Research (CCARR) Group

BACKGROUND: The Centers for Medicare & Medicaid Services (CMS) Patient-Reported Outcome-Based Performance Measure (PRO-PM) mandates collection of patient-reported outcome measures (PROMs) after total knee arthroplasty (TKA) and defines success as achieving substantial clinical benefit (SCB) on the Knee injury and Osteoarthritis Outcome Score for Joint Replacement (KOOS-JR) of ≥ 20 points. As TKA increasingly shifts to outpatient care and patient complexity varies by care setting, a single universal threshold may misclassify improvement. This study estimated specific KOOS-JR SCB thresholds in separate Medicare inpatient and outpatient cohorts and compared the performance of these thresholds between settings.

METHODS: A retrospective analysis of prospectively collected data on 7,926 Medicare beneficiaries ≥ 65 years who underwent primary elective TKA from 2016 to 2022 was performed. Patients were stratified by care setting into inpatient (n = 2,812) and outpatient (n = 5,114) cohorts. The SCB thresholds on KOOS-JR were derived using an anchor-based approach with the Veterans RAND 12-Item, "Compared to one year ago, how would you rate your physical health now?" as the anchor. Receiver operating characteristic analysis with Youden's index identified setting-specific cut points.

RESULTS: The KOOS-JR SCB threshold for the inpatient Medicare cohort was 29.3 points (Youden's index 0. [...]

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Risk and Benefit Profile with Concurrent Nonsteroidal Anti-Inflammatory Drugs and Direct Oral Anticoagulants Following Total Knee Arthroplasty.

Paul BR, Verhey JT, Haak GM, Braithwaite CL, Tarabichi S, Christopher ZK, Spangehl MJ, Bingham JS

BACKGROUND: Nonsteroidal anti-inflammatory drugs (NSAIDs) are commonly prescribed after total knee arthroplasty (TKA) for analgesia and inflammation control in conjunction with direct oral anticoagulants (DOACs) for venous thromboembolism (VTE) prophylaxis in high-risk patients. In non-orthopaedic populations, concurrent NSAID-anticoagulant increases bleeding risk, but post-TKA safety data remain limited.

METHODS: Using a national claims database (2010 to 2023), we identified patients undergoing primary TKA who received DOAC-only VTE prophylaxis for $\geq$ three days. Patients co-prescribed NSAIDs were propensity score-matched 1:3 to DOAC-only patients based on demographics, comorbidities, and preoperative anticoagulant use. The matched cohort included 12,733 patients who received DOAC plus NSAID therapy and 37,127 who received DOAC alone. Outcomes included 90-day medical and surgical complications, two-year implant-related complications, and arthrofibrosis-associated outcomes. Subanalyses evaluated outcomes by NSAID type and prescription duration.

RESULTS: Combination therapy was associated with increased wound dehiscence (odds ratio (OR) 1.49, 95% confidence interval (CI) 1.20 to 1.87), surgical site infection (OR 1.71 [1.25 to 2.34]), and periprosthetic joint infection (OR 1.24 [1.01 to 1.52]), without differences in 90-day bleeding events ($P > 0.05$). At two years, NSAID plus D [...]

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Complications of Ceramic-on-Ceramic Total Hip Arthroplasty with Large Heads: 10-Year Minimum Follow-Up.

Lee DH, Park JW, Jung WH, Kim TY, Ha YC, Koo KH, Lee YK

BACKGROUND: The fourth-generation ceramic bearings allow the use of larger femoral heads, but inevitably create thinner liners, raising concerns regarding liner fracture and long-term biological responses. We extended our study to evaluate late dislocation, ceramic-related complications, and long-term clinical and radiographic outcomes more than 10 years after ceramic-on-ceramic (CoC) total hip arthroplasty (THA).

METHODS: We conducted a prospective multicenter study of cementless THA using a 32/36-mm ceramic bearing. From April 2010 to February 2012, 246 patients (274 THAs) were enrolled. We evaluated ceramic fracture, noise, clinical results, radiological changes, and survival rate at a minimum of 10-year follow-up. In this study, 178 patients (100 men and 78 women, 201 hips) who had a mean age of 51 years (range, 18.9 to 77.6) at surgery were followed for a mean of 12.3 years (range, 10 to 14.6) with computed tomography (CT) scans in 47 hips (23%, 47 of 201). Kaplan-Meier survival analysis was performed with revision and reoperation as endpoints.

RESULTS: In addition to previously reported three dislocations, the new one sustained anterior instability due to lumbar degenerative kyphosis (LDK) associated with posterior impingement. Ceramic fracture did not occur in any patient. The mean Harris Hip Score was 87.0 ± 11.5 points at the latest follow-up. New focal osteolysis wa [...]

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Disparities in Preoperative Weight Loss and Obesity Treatment Before Total Joint Arthroplasty.

Seward MW, Grimm JA, Bedard NA, Hannon CP, Berry DJ, Abdel MP

BACKGROUND: Institutions often implement body mass index (BMI) cutoffs for offering total hip (THA) and knee (TKA) arthroplasty. The goals of this study were to determine disparities in preoperative obesity prevalence, obesity treatment utilization, and weight loss efficacy.

METHODS: Among 21,038 primary THAs and 23,726 TKAs performed between 2002 and 2019, we identified 6,128 patients who had preoperative BMIs ≥ 30 measured one to 24 months before surgery, including 4,953 with available Area Deprivation Index (ADI) information. The mean age was 67 years (range, 19 to 97) with 55%

women. The mean BMI was 36 (range, 20 to 69). The ADI was used to analyze socioeconomic status. Univariable and multivariable analyses evaluated the use of nutrition services or bariatric surgery, the likelihood of weight loss, and ADI's potential association with weight loss.

RESULTS: A few patients received preoperative weight loss medications (0.2%), nutrition services (1.6%), and/or bariatric surgery (2.6%). Compared to the best (least deprived) ADI quartile 1, the worse ADI quartiles 3 and 4 were associated with an increased odds of BMI ≥ 40 preoperatively (odds ratio (OR) 1.4, $P = 0.01$ and OR 1.9, $P < 0.01$, respectively). Women were associated with an increased odds of BMI ≥ 40 preoperatively (OR 1.6, $P < 0.01$) and were more likely to receive bariatric surgery (OR = 2.3, $P < 0.01$). Women had I [...]

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A Proposed Alternative Sampling Strategy for Centers for Medicare & Medicaid Services Arthroplasty Patient-Reported Outcome Performance Measure Compliance.

Lyman S, Chin A, Alschuler DM, Brill CO, Fontana M, MacLean C, McLawhorn A

INTRODUCTION: The Centers for Medicare & Medicaid Services (CMS) mandate reporting of Hip disability and Osteoarthritis Outcome Score for Joint Replacement (HOOS, JR.) and Knee injury and Osteoarthritis Outcome Score for Joint Replacement (KOOS, JR.) to support a hospital-level patient-reported outcome performance measure (PRO-PM). The current CMS requires matched pre- and postoperative PROMs for $\geq 50.0\%$ of all eligible patients, regardless of hospital volume. This may yield unreliable performance estimates for low-volume centers and potentially biased and unnecessarily large estimates for large hospitals. The purpose of this study was to determine the smallest sample required to produce statistically valid PRO-PM estimates.

METHODS: We calculated the minimum number of completed PROMs needed in order to achieve a 95% confidence interval (CI) width 10.0% ($\pm 5.0\%$) for hospitals with five, 50, 500, and 5,000 eligible total hip arthroplasty (THA) and total knee arthroplasty (TKA) cases annually, assuming a conservative substantial clinical benefit (SCB) achievement of 50.0%. These values were compared with current CMS requirements. Sensitivity analyses explored the effects of confidence interval width, hospital volume, and frequency of SCB attainment on required sample sizes.

RESULTS: The CMS's volume-proportional rule requires three, 25, 250, and 2,500 patients for hospitals wit [...]

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First Revision Total Hip Arthroplasty for Instability: 10-Year Outcomes and Risk Factors Associated with Re-Revision.

Yapp LZ, Howard LC, Greidanus NV, Garbuz DS, Masri BA, Neufeld ME

BACKGROUND: Recurrent instability remains a leading cause of failure after revision total hip arthroplasty (rTHA). This study assessed the 10-year survivorship of the first rTHA when undertaken for instability and identified factors associated with failure.

METHODS: We retrospectively identified all consecutive rTHA performed for instability at our institution between 2000 and 2022. Of the 678 rTHA identified, 284 (41.9%) were the first rTHA and included in the study. Re-revisions and oncology cases were excluded. The mean age at revision was 65 years, and 63.2% were women. The mean follow-up was 11.4 years (range, two to 22). A total of 84 (29.6%) hips underwent at least one re-revision. Survival free from instability-related and all-cause re-revision at 10 years was calculated using Kaplan-Meier analysis. Multivariable Cox proportional hazard regression modeling was used to assess factors associated with 10-year survival.

RESULTS: The 10-year survivorship was 82.2% for instability and 73.6% for all-cause re-revision. Dual mobility (DM) components had higher 10-year instability-free survival than both standard bearings and constrained liners ($P = 0.04$). However, when considering 'large' (≥ 36 mm) femoral heads, there was no difference in the revision rate when compared to DM components ($P = 0.59$). Constrained liners were associated with a higher risk of revision for instabil [...]

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Jumbo Femoral Heads in Total Hip Arthroplasty: Improved Early Stability Without Added Complications.

Kim BI, Cheng R, Grabov E, Debbi EM, Vigdorichik JM, Mayman DJ, Shen TS

BACKGROUND: Larger femoral heads have been proposed to reduce dislocation risk following total hip arthroplasty (THA). While the benefit of increasing head size from 28 or 32 mm to ≥ 36 mm is well established, it remains unclear whether jumbo (40 or 44 mm) femoral heads confer additional stability over traditional large (36 mm) heads. This study aimed to determine whether jumbo heads improve stability or alter early complication rates.

METHODS: We identified 6,413 primary THAs with large-diameter femoral heads (36-, 40-, or 44-mm) implanted at a single institution between February 1, 2016, and December 31, 2022. Cases were stratified by femoral head diameter into the traditional large-head (36 mm, $n = 5,897$) and jumbo-head (40 or 44 mm, $n = 516$) groups. The median follow-up was 2.4 years (range, 0.1 to 9.8; interquartile range [IQR], 1.3 to 4.1). The primary outcome was prosthetic hip dislocation at the last clinical follow-up. Time-to-event analyses were performed using multivariable Cox models with surgeon-level random effects, adjusting for patient-specific factors, year of surgery, surgical approach, and robotic assistance.

RESULTS: In the full follow-up period, jumbo heads were not associated with a lower adjusted dislocation risk (hazard ratio (HR): 0.55 [95% confidence interval (CI): 0.13 to 2.34], $P = 0.139$) when compared with 36 mm heads. When analysis was truncated [...]

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Dogma Under the Knife: Challenging the Status Quo in Primary Total Knee Arthroplasty.

Padgett BM, Terhune EB, Chalmers BP, Pagnano MW, Hannon CP

INTRODUCTION: Primary total knee arthroplasty (TKA) is among the most successful procedures in orthopaedic surgery, reliably providing pain relief, functional improvement, and durable implant survivorship. Despite these achievements, approximately 10 to 20% of patients report persistent dissatisfaction following primary TKA, even in the absence of technical failure. This disconnect between implant longevity and patient-perceived success has prompted renewed examination of the foundational principles that have historically guided TKA practice. Advances in implant design, biomaterials, surgical technology, and a changing patient population have challenged long-standing dogma related to alignment philosophy, fixation strategy, bearing surface selection, and patellar resurfacing.

METHODS: This symposium manuscript reviews contemporary evidence addressing the evolving paradigms in primary TKA, with a focus on individualized alignment strategies, the growing role of cementless fixation, expanded bearing design options, and selective approaches to patellar resurfacing.

RESULTS: In each of these areas, the goal is not to promote a single new standard, but rather to provide an evidence-based framework to support thoughtful, patient-specific decision-making.

CONCLUSION: As the field continues to evolve, moving beyond rigid dogma toward personalized reconstruction strategies may repres [...]

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Revision Total Joint Arthroplasty at the Ambulatory Surgery Center: A Single Institutional Experience.

Acuña AJ, Potluri AS, Yadav AS, Alfonsi S, Forlenza EM, Della Valle CJ

INTRODUCTION: The safety and efficacy of outpatient primary total joint arthroplasty (TJA) in appropriately selected patients have been established. As rates of revision TJA (rTJA) continue to increase, there is growing interest regarding the complications and revision incidence among procedures performed in the outpatient setting. The purpose of this study was to report outcomes of revision TJA procedures performed at an ambulatory surgery center (ASC).

METHODS: A retrospective case series review of 181 patients undergoing aseptic rTJA at a free-standing ASC between January 1, 2017, and July 1, 2024, at a single large academic referral center was performed. Patients who did not have a minimum 1-year follow-up were excluded ($n = 28$). Our primary outcome was the incidence of complications within 90 days of rTJA, with our secondary outcome as the incidence of re-revision at final follow-up. A majority of patients underwent modular component exchanges (75.7%), with instability being the most common revision etiology (71.8%). The mean follow-up was 3.4 years (range, 1.1 to 8.6). A Kaplan-Meier survival analysis was performed to identify survivorship free of all-cause re-revision at two and five years postoperatively.

RESULTS: There was one patient (0.6%) who developed a DVT within 90 days postoperatively. There were no other medical complications seen among included patients. The [...]

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The "Distal Femoral Replacement Downgrade": Technique of Using Bicondylar Femoral Cones with a Hinge Total Knee Arthroplasty Revision in Patients Who Have Massive Distal Femoral Bone Loss as an Alternative.

Chalmers BP, Afzal S, Bhatti P, Anatone AJ, Blevins JL, Debbi EM, Sculco PK, Gausden EB

BACKGROUND: Massive femoral bone loss with collateral insufficiency in revision total knee arthroplasty (TKA) is often managed with distal femoral replacement (DFR), but this procedure sacrifices bone stock and carries notable risks. We described outcomes of a bone-preserving alternative (the "DFR downgrade") that pairs bicondylar, metadiaphyseal-engaging femoral cones with a hinge construct.

METHODS: We performed a retrospective review of consecutive patients who underwent revision TKA using a "DFR downgrade" technique for severe femoral bone loss between February 2016 and July 2024. There were twenty patients (mean age, 69 years (range, 58 to 86); 65% men) included, with a mean follow-up of 14.5 months (range, 0.9 to 43.7). Indications were aseptic loosening in 75%, infection in 10%, ligamentous instability in 10%, and arthrofibrosis in 5%. Demographics, Anderson Orthopaedic Research Institute (AORI) classification, operative details, postoperative complications, and clinical and radiographic outcomes were collected. The Kaplan-Meier method was used to estimate reoperation-free and re-revision-free survivorship. Femoral bone loss was classified as AORI IIA in 10%, IIB in 30%, and III in 60% of cases.

RESULTS: Survivorship free from revision for aseptic loosening was 100% at one and two years. Survivorship free from any re-revision was 85.1% at one and two years. Survivorshi [...]

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Does the surgical sequence of lumbar spine fusion and total hip arthroplasty affect the dislocation and revision risk? A meta-analysis and systematic review.

Liu F, Jia C, Wu X, Zhou Y

BACKGROUND: Patients who underwent primary total hip arthroplasty (THA) after lumbar spine fusion (LSF) had a higher incidence of complications compared with those who had not previously undergone LSF. This study aimed to determine whether the timing of THA before or after LSF impacts the incidence of THA complications.

METHODS: Following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, we searched PubMed, Web of science, Embase, and Cochrane Library from their inception until 1 October 2025. Data from all eligible published studies meeting the inclusion criteria were extracted and subsequently analyzed using a random-effects model to calculate the rates of all-cause revision, dislocation, periprosthetic joint infections, and aseptic loosening.

RESULTS: Our analysis included 10 studies involving 27,605 patients who underwent THA followed by LSF and 32,002 patients who received LSF prior to THA. There are no significant differences in the rates of dislocation (odds ratio [OR] 1.19, 95% confidence interval (CI) 0.73-1.86, $p = 0.45$), periprosthetic joint infections (OR 0.91, 95% CI 0.52-1.62, $p = 0.76$), and aseptic loosening (OR 0.89, 95% CI 0.74-1.08, $p = 0.23$) between the two surgical sequences, while for all-cause revision, the OR was (OR 1.01, 95% CI 0.70-1.47, $p = 0.94$), which reached statistical significance.

CONCLUSIONS: Compare [...]

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Letter to the Editor Regarding "Proton Pump Inhibitors Are Associated With Increased Risk of Site-Specific Nonunion After Open Reduction Internal Fixation".

Ayas O, Acar A

[PubMed](#) · [DOI](#)

Response to "Competing-Risks Re-estimation of Five-Year Second Hip Fracture Risk: Accounting for Death as a Competing Event".

Konda SR

[PubMed](#) · [DOI](#)

Response Letter to the Editor Regarding Article "Orthopaedic Trauma Association Annual Meetings Consistently Provide Academic Value for Attendees".

Greenhill DA, Lundy D, Imam N, Jain N

[PubMed](#) · [DOI](#)

Letter to the Editor Regarding Article "Orthopaedic Trauma Association Annual Meetings Consistently Provide Academic Value for Attendees".

Blomberg JR

[PubMed](#) · [DOI](#)



Does the tendon retract in flexor hallucis longus muscle laceration? A cadaveric study.

Aydin M, Mahin Y, Yasar H, Çiçek F, Ceranoğlu FG, Mert A, Çınarolu S

INTRODUCTION: Flexor hallucis longus (FHL) tendon injuries are common, particularly in open injuries caused by sharp objects. However, data regarding FHL tendon retraction following hallux-level FHL tendon lacerations remain limited. This study investigated the retraction and shortening of the FHL tendon using cadaveric models.

MATERIALS AND METHODS: Six fresh-frozen below-knee amputated cadaveric feet (mean age: 65 years, range: 62-88) were utilized. FHL tendons were dissected, and lacerations were simulated at the hallux level. Tendons were subjected to various tensile forces (20-120 N) using a mechanical device. Force was measured in Newtons via a loadcell, and retraction distance was recorded in millimeters using an encoder. Data were summarized as mean $\pm$ standard deviation (SD), median, and quartiles.

RESULTS: The FHL tendon retracted gradually as applied force increased. An average retraction of 7 mm was measured at 20 N, reaching 21 mm at 60 N. Beyond 60 N (> 21 mm retraction), the other four toes also flexed. This occurs due to the anatomical connection between the FHL and flexor digitorum longus (FDL) tendons.

CONCLUSIONS: Only slight retraction was observed in hallux-level FHL tendon lacerations. At most, the findings suggest that distal FHL lacerations may exhibit limited proximal retraction under the tested conditions, which may have implications for surgical pla [...]

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Early-onset rice body synovitis of the knee in children under three years: a case series and review of the literature.

Zhu L, Liu C, Liu Y, Mao B, Huang Y, Wen J, Liu Y, Qian T

OBJECTIVE: Rice body synovitis reports have been rarely reported in children under three years of age. We aimed to explore the clinical characteristics and treatment experience of rice body-like synovitis in children's knees.

METHODS: We retrospectively analyzed the clinical and imaging data of 4 cases (5 knees) of rice body-like synovitis in children treated at our hospital from January 2014 to December 2021. Among the patients, 1 was male and 3 were female, with an average age of 22.5 months. The average length of symptoms before admission was 1.86 months. Clinical presentations included swelling and limping, with no fever or rash.

RESULTS: Three patients with unilateral onset and one patient with bilateral onset underwent unilateral arthroscopic excision of synovial lesions. All 4 patients had pathological findings consistent with chronic inflammation, leading to a postoperative diagnosis of juvenile idiopathic arthritis (JIA). Postoperatively, all patients received standardized pharmacological treatment and were followed for 24 to 48 months in collaboration with the rheumatology department. None of the 4 knees undergoing surgery experienced recurrence, and the symptoms of the knee treated conservatively resolved spontaneously. In the last follow-up session, MRI and ultrasound showed only mild synovial thickening, with effusion and loose bodies disappearing.

CONCLUSION: R [...]

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Extravertebral temperature exposure during spinal radiofrequency ablation: an experimental surrogate assessment of neural injury risk.

Weigert A, Hoffmann RT, Büttner A, Zaccaria R, Wegener V, Holzapfel BM, Wegener B

BACKGROUND: Radiofrequency ablation (RFA) has become an important minimally invasive option for the treatment of painful spinal tumors and benign bone lesions. While its effectiveness within the vertebral body is well established, there is ongoing concern about unintended heat exposure of nearby neural structures. Experimental data describing how heat propagates beyond the vertebra during spinal RFA are still limited. This experimental in vitro study aims to assess the extravertebral temperature exposure during spinal radiofrequency ablation as a surrogate parameter for potential neural injury risk under controlled conditions.

METHODS: This experimental in vitro study was performed on four fresh-frozen human lumbar spines. Radiofrequency ablation was carried out using a commercially available system and standard clinical protocols. Two anatomical configurations were examined: ablation within the vertebral body and ablation within the pedicle. Temperatures were measured at predefined locations outside the vertebra, including the spinal canal and cortical boundaries, using multiple thermocouples. To aid interpretation of radiofrequency-related measurement artifacts, an additional control setup using an electrothermal heating system without radiofrequency energy was included. Temperature data were evaluated descriptively.

RESULTS: In the control experiments, temperature profiles [...]

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Giant cell tumor of bone: exploring HtrA1 as a potential predictor of recurrence.

Mirioglu A, Dalkir KA, Gokmen MY, Bagir M, Ates KE, Deveci MA, Gönlümen G, Ozbarlas HS

INTRODUCTION: Giant cell tumor of bone (GCTB) is a locally aggressive tumor with a considerable risk of recurrence, but reliable predictive markers are limited. High-temperature requirement A serine peptidase 1 (HtrA1) has been implicated in the progression of several cancers and may also play a role in GCTB recurrence.

MATERIALS AND METHODS: This retrospective study included 34 patients with GCTB treated with curettage between 2002 and 2016. HtrA1 expression was evaluated separately in giant cells (GC) and mononuclear cells (MNC) using immunohistochemistry. Based on high or low expression in each cell type, patients were categorized into four expression patterns: High GC / High MNC, High GC / Low MNC, Low GC / High MNC, and Low GC / Low MNC. Clinical data, including recurrence status and time to recurrence, were analyzed.

RESULTS: The High GC / Low MNC group showed the highest recurrence rate at 71.4%, compared with 28.6% in the High GC / High MNC group and 31.6% in the Low GC / Low MNC group. No recurrence was observed in the Low GC / High MNC group, which included only one patient. The High GC / Low MNC group showed a non-significant trend toward higher recurrence compared with all other groups combined, with a similar trend

toward shorter recurrence-free survival.

CONCLUSION: HtrA1 expression patterns in GCTB may provide preliminary insight into recurrence risk. Although [...]

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Injury

Volume 57, Issue 8 · June 2026 · 11 new articles

More than meets the eye: the continuing importance of tertiary trauma survey even in moderate (ISS 9-15) trauma.

Knight C, Burke JR, Hind S, Smith SR

INTRODUCTION: Tertiary Trauma Surveys (TTS) are designed to identify injuries which are not recognised during the primary and secondary surveys. These injuries occur in up to 40% of patients and are commonly referred to in the literature as "missed", we prefer the term "unrecognised" as this term does not infer the failure of primary and secondary survey, rapid assessments designed to identify life and limb threatening injuries. The incidence and definition of unrecognised injuries varies across studies, and several risk factors have been identified internationally. The aim of our study was to investigate the sensitivity and specificity of TTS and the incidence and risk factors for unrecognised injuries in a large UK Major Trauma Centre (MTC).

METHODS: Data was prospectively collected for all major trauma patients admitted to our MTC over six months. All patients over 16 years with an ISS $\geq$ 9 were included. Patients with a length of stay of less than 24 h or palliated were excluded. All patients were examined using a specific TTS protocol. Patient records were reviewed one year post admission to identify any injuries diagnosed after discharge.

RESULTS: A total of 293 patients were included. Unrecognised injuries were identified at TTS in 18.8% (55/293) patients, most commonly affecting the face or head and eyes. Patients with unrecognised injuries had a higher admission ISS (p [...])

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CTA-detected incidence and descriptive fracture patterns of arterial injury in 539 surgically treated tibial plateau fractures.

Valderrama J, Narváez F, Cancino J, Raffo JF, Bosselin R, Zamora F, Siegel S, Scheu M et al.

BACKGROUND: Tibial plateau fractures (TPF) are complex injuries associated with significant soft tissue and periarticular complications. Although vascular injuries are well described in lower limb trauma, their incidence in tibial plateau fractures has not been clearly established.

METHODS: Retrospective cohort study including patients with surgically treated TPF at a level I trauma center between 2016 and 2024. CTA was selectively performed based on clinical judgment. Post hoc descriptive risk scenarios were constructed to organize observed fracture patterns. Analysis was primarily descriptive.

RESULTS: 539 surgically treated TPF were included. Mean age was 43.7 years (18-75), 73.84% were male. Left knee was affected in 54.92%. High-energy trauma accounted for 70.87%. The CTA yield was 3.66% (6/164), corresponding to an overall observed incidence of 1.11% (6/539). Certain fracture patterns and clinical scenarios were more frequently observed among cases with arterial injury; however, these observations are based on very few events and should be interpreted cautiously as descriptive findings only.

CONCLUSION: The CTA-detected incidence of arterial injury in this series of surgically treated tibial plateau fractures was 1.11%. Certain fracture patterns and clinical contexts were observed among the small number of cases with arterial injury; however, these findings are descrip [...]

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Diagnostic value of serum troponin in stable patients at risk for blunt cardiac injury.

Cleary HL, Bernard MD, Davenport DL, Bernard AC

INTRODUCTION: There are no gold standard criteria for diagnosing blunt cardiac injury (BCI). While the combination of electrocardiogram (ECG) and serum troponin is considered the most sensitive screening method for those being considered for discharge, little evidence exists regarding the diagnostic value and clinical utility of initial and serial serum troponin levels in stable patients who are being admitted with a possible BCI. Therefore, this study seeks to determine if troponin level is an independent predictor of adverse cardiac events in stable, admitted patients at risk for BCI.

METHODS: This was a five-year retrospective study using the trauma database at a University Level I Trauma Center. The study population included adult trauma patients presenting with a physician diagnosis of BCI or a sternal fracture who met prespecified stability criteria (SBP $\geq$ 90 mmHg, HR < 110bpm, shock index < 1, GCS $\geq$ 14). The data collected included all troponin values and ECG interpretations as well as echocardiogram interpretations, when performed. A patient was classified as having an adverse cardiac event if they were diagnosed with a new arrhythmia requiring treatment, had cardiac surgery, or suffered cardiac-related mortality. Sensitivity, specificity, positive and negative likelihood ratios, and diagnostic odds ratios were calculated to analyze the diagnostic performance of the di [...]

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Risk Factors for Failure of Repair of Diaphyseal Tibial Nonunions: A Retrospective Cohort Study of 108 Patients.

Fan S, Wagner RK, Lehle CH, Musick AN, Gregg AT, Janssen SJ, Wijffels MME, Stenquist DS et al.

OBJECTIVES: To determine rates and risk factors for additional nonunion surgery and final radiographic healing following diaphyseal tibial nonunion repair.

METHODS: All consecutive patients aged $\geq$ 18 years who underwent diaphyseal (AO/OTA 42) tibial nonunion repair with (exchange) intramedullary nailing (IMN) or open reduction and internal fixation (ORIF) between 2001 and 2024 with $\geq$ 6 months follow-up were included. The primary outcome was the rate of additional nonunion surgery. The secondary outcome was radiographic healing (defined as Modified Radiographic Union Score for Tibia [mRUST] $\geq$ 11) at final follow-up. Multivariable logistic regression was used to determine the association of patient, nonunion, and treatment characteristics with additional nonunion surgery.

RESULTS: A total of 108 patients were included (70% male, median age 42 years [IQR 28-53]). Sixty patients (56%) were treated with IMN, of whom 54 (50%) underwent exchange IMN, and 48 (44%) were treated with ORIF. Autograft was used in 39 patients (36%) and bone graft substitutes in 21 (19%). Twenty-seven patients (25%) required additional nonunion surgery, of whom 5 (4.6%) ultimately underwent amputation. In multivariable analysis, current smoking (OR: 4.28, p = 0.015), initial open fracture (OR: 3.94, p = 0.045), and presence of a bone defect (OR: 9.20, p < 0.001) were associated with increased odds of addition [...]

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Superior survival, hemodynamics, and metabolic recovery using PEG-20k-based low volume resuscitation in a polytrauma hemorrhagic shock model in rats.

Khoraki J, Payne C, Lust D, Mangino MJ, Liebrecht LK

BACKGROUND: Hemorrhagic shock is a major cause of preventable civilian and military trauma death. Although whole blood and blood products remain the standard of care, field-stable alternatives are needed for resource-limited environments and supply-chain disruptions. Blood-based resuscitation may also fail to optimally restore microvascular perfusion despite improved macrohemodynamics, contributing to ongoing ischemia-reperfusion injury, dysfunctional molecular cell dynamics, and capillary cell swelling. Prior controlled hemorrhage studies in rats and swine demonstrated superior survival, mean arterial pressure (MAP), and lactate clearance with polyethylene glycol 20,000 (PEG-20k) compared with crystalloid, colloid, and whole blood controls. These beneficial effects are attributed to the capacity of PEG-20k to augment water transfer from swollen ischemic cells into the vascular space, thereby decompressing capillary beds and restoring effective tissue perfusion. This study evaluated PEG-20k in a severe polytrauma rat model with combined controlled and uncontrolled hemorrhage.

METHODS: Adult male Sprague-Dawley rats (250-400-g) underwent a midline laparotomy, 75% tail amputation, and dual splenic transections. The spleen bled freely into the closed abdomen (uncontrolled, non-compressible hemorrhage), while the tail was intermittently clamped (controlled, compressible hemorrhage [...])

The forequarter lateral implosion injury: A diagnostic paradigm and Ch-Sh (ch■-sh■) classification.

George A, Paull TZ, Winter JD, Levine J, Cole PA

BACKGROUND: The lateral impaction force against the shoulder often causes multiple lesions to the forequarter including the chest wall. The purpose of this study is to describe the injury patterns and associated pathology in patients with combined chest wall and shoulder girdle injuries- forequarter lateral implosion injury and organize the findings into a classification framework to predict morbidity.

METHODS: Three hundred fifty-four consecutive patients with two or more rib fractures and ipsilateral shoulder girdle injury from a Level 1 trauma center, between 2018-2022, were retrospectively reviewed. Ch-Sh is pronounced ch■ sh■. Chest injuries were grouped into three categories: Ch1: 2-4 fractured ribs, Ch2: 5 + fractured ribs, or Ch3: 2 + fractured ribs with contralateral rib fractures and/or sternal fracture. Shoulder girdle injuries were grouped into: Sh1: clavicle or acromion fracture, and/or AC/SC joint dislocation, Sh2: scapula and/or proximal humerus fracture, or Sh3: combination of Sh1 and Sh2 injuries (clavicular chain and scapula/humerus). There were 9 total subgroups. Demographics, injury characteristics, associated injuries, and outcomes were recorded. Inter and intraobserver reliability of the classification system was studied.

RESULTS: As chest wall injuries scaled from Ch1 to Ch3, there were significantly higher rates of ICU admission, pneumothorax, pulmonar [...]

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The fate of stage I Masquelet cement spacers in long bone reconstruction for acute fractures.

Dekle JK, Sisk CK, Litten RM, Alcaide DM, McIlwain RN, Wilson AL, Manson HC, Spitler CA et al.

INTRODUCTION: The Masquelet technique is widely used for reconstruction of segmental long bone defects following acute fractures; however, not all patients progress to the second-stage bone grafting procedure. The purpose of this study was to characterize Stage II completion following Stage I Masquelet cement spacer placement for acute long bone fractures, compare characteristics and outcomes between patients who underwent Stage I only versus completed Stage II reconstruction, and identify factors associated with progression to Stage II bone grafting.

METHODS: A retrospective review was conducted of adult patients who underwent Masquelet reconstruction for acute long bone fractures at a Level I trauma center between 2014 and 2024. Patients were stratified by completion of Stage II reconstruction. Demographic, injury-related, surgical, and clinical outcome variables were collected. Multivariable logistic regression was performed to identify factors independently associated with Stage II completion.

RESULTS: A total of 156 patients met inclusion criteria, of which 48 (30.8%) underwent Stage I only and 108 (69.2%) completed both stages. Patients who remained at Stage I underwent fewer orthopaedic operations (3.2 vs. 4.2, $p = 0.001$) and initial bony defect size did not differ between groups (84.1 vs. 78.8 mm, $p = 0.423$). Among patients with at least 6 months of follow-up, rates o [...]

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Rethinking the Lisfranc injury: Rationale for and development of a new classification system for Lisfranc injuries.

Cawley WCF, Hwang P, Mittal R, Zochowski Y, Darwish I, O'Keefe R, Grogin J, Sivakumar B et al.

BACKGROUND: The Lisfranc joint comprises the articulations between the cuneiforms and bases of the metatarsals and is susceptible to a variety of complex injuries which are frequently missed or diagnosed late. This can result in morbidity with poor outcomes, and renders it challenging to standardise management and allow prognostication. Existing classification systems, including the Myerson-Hardcastle (MHC) and Nunley & Vertullo classifications, do not consider the propagation of force through the midfoot, which is key to identifying structures which may require stabilization. They also demonstrate poor clinical utility due to poor correlation with outcomes. This study aimed to develop a new classification system for Lisfranc injuries using data from a large single-surgeon

case series from a tertiary level trauma hospital in Australia. The primary outcome measures for this study were inter- and intra-observer reliability.

METHODS: The Suthersan classification was developed to address shortfalls in current classification systems. Patient records were manually searched for Lisfranc injuries. Direct 'crush' Lisfranc injuries were excluded due to unpredictable injury patterns. Indirect Lisfranc injuries were included in analysis. Two orthopaedics foot and ankle surgeons and one hand surgeon reviewed patient radiographs and classified them using the MHC and Suthersan classification [...]

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Injury and trauma care training in global health: A scoping review of costs and cost-effectiveness across resource settings.

Phan HT, Nguyen BT, Zhao T, Phung LT, Keakabetse TR, Dinh TTN, Nelson MR, Skinner M et al.

INTRODUCTION: Injury is a leading cause of death and disability worldwide, with a disproportionate burden in low- and middle-income countries, where health resources are constrained and funding decisions are highly cost-sensitive. Injury and trauma care training programs have demonstrated their effectiveness in enhancing the capacity of care providers to treat injured patients, but evidence on their costs and cost-effectiveness remains limited and fragmented. This scoping review aims to map and summarise current literature on costs and cost-effectiveness of injury and trauma care training across different resource settings.

METHODS: We conducted a systematic search on PubMed, MEDLINE Complete, Embase via Ovid, Web of Science, EconLit and Emcare. Papers published in English that reported implementation costs or cost-effectiveness of trauma care training programs were included. Cost data were converted and presented as USD 2024.

RESULTS: Twenty-eight studies were included, predominantly in low- and middle-income countries. Implementation costs varied considerably, starting from USD 3.70 (Lay-First-Responder) to USD 1433.5 (Advanced Trauma Life Support). Foreign instructor involvement was a particularly significant cost driver, accounting for 38-85% of total implementation costs in some programmes. Training-of-trainers and cascading models were effective in reducing costs, with [...]

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Do reduced screws have a significant impact on fixation in medial open wedge high tibial osteotomy? A comparative fatigue and stability analysis.

Özkaya M

INTRODUCTION: Medial opening wedge high tibial osteotomy generally employs T-shaped plates with three or more proximal screws to ensure stabilization. This study aims to evaluate the biomechanical necessity of the standard three screw proximal configuration of the T-shaped plates and investigates whether fixations with a reduced number of proximal screws provide sufficient stability and fatigue life under diverse loading scenarios using finite element analysis.

MATERIALS AND METHODS: Three fixation models were analyzed, a standard T-shaped plate with three proximal screws in the first row and two modified models with two proximal screws in the first row with or without an empty hole. Models were subjected to physiological axial loading, centrally, anteriorly, or posteriorly applied in the sagittal plane, to measure total deformation, displacement of the tibial plateau, fixation stiffness, and von-Mises stress. Furthermore, a fatigue analysis was conducted to evaluate the endurance of fixations under these varying loading conditions.

RESULTS: The reduction in proximal screws resulted in a marginal stiffness decrease of less than 2%, with values remaining above 7700 N/mm under centric loading. Total deformation and tibial plateau displacement showed negligible differences between two screw and three screw configurations. In addition, all peak stresses remained well below the [...]

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Moving beyond mortality measurements after road traffic injuries: Ten-year death risk after road traffic injuries.

Cherchiglia ML, Mougin V, Gray A, Wool EE, Malta DC, Melo APS, de Assis Acúrcio F, Junior AAG et al.

Road traffic injuries (RTIs) represent a significant global public health issue, particularly in low- and middle-income countries like Brazil. Despite the high mortality rates and associated costs, there is limited comprehensive literature addressing long-term mortality resulting from RTIs. Most research tends to focus on rates and hospitalization figures while neglecting the connection between hospitalizations and causes of death, partly due to the challenges of follow-up studies. This study employed record-linkage methodology to create a database that includes both hospitalization and mortality records. The aim was to estimate all-cause and cause-specific relative risks (RR) of death among inpatients due to road traffic accidents, compared to those hospitalized for other reasons, over a time span of three months to ten years. A retrospective cohort study examined hospitalization data for RTIs and mortality from 2000 to 2015 using Brazil's Unified Health System (Sistema Único de Saúde - SUS in Portuguese), excluding any obstetrics-related admissions and deaths, and calculated RR stratified by sex and age over 15 years old. The findings indicated a significantly higher mortality risk of 3.23 (95% Uncertainty Interval: 3.08- 3.39) for individuals previously hospitalized for RTIs compared to others during follow-up. Males aged 15- 29 and 30- 59, especially those with motorcycle, [...]

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Denis pattern-dependent biomechanical behavior of posterior trans-ilial plate fixation compared with bilateral triangular osteosynthesis in unilateral sacral fractures: a finite element study.

Yang J, Böker KO, Li C, Schilling AF, Ritter Z, Acharya M, Lehmann W

OBJECTIVE: Posterior trans-ilial plate osteosynthesis (TPO) is commonly used for the stabilization of unilateral vertically unstable sacral fractures. However, its biomechanical performance across different Denis fracture patterns remains unclear. This study aimed to investigate the fracture pattern-dependent biomechanical behavior of posterior trans-ilial plate osteosynthesis (TPO) in unilateral vertically unstable sacral fractures, using bilateral triangular osteosynthesis (BTO) as a high-stability reference construct.

METHODS: A validated three-dimensional finite element model of the lumbopelvic complex was developed to simulate unilateral vertically unstable sacral fractures corresponding to Denis type I, II, and III patterns. Two posterior fixation strategies, TPO and BTO, were constructed for each fracture type. Seven physiological loading conditions were applied, including standing, flexion, extension, axial rotation, and lateral bending. Von Mises stress distribution and fracture displacement were evaluated. Fracture micromotion was quantified by calculating relative displacement changes between paired points along the fracture gap.

RESULTS: When interpreted relative to the high-stability BTO reference construct, TPO showed fracture pattern-dependent variation in fracture displacement. The difference was most pronounced in Denis type I fractures and progressively decreased [...]

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Factors associated with in-hospital mortality in necrotising soft tissue infections. a multicentre retrospective cohort study.

Podda M, Ceresoli M, Viridis F, Rosa F, Pilia T, Murzi V, Dessì A, Tedesco S et al.

PURPOSE: Necrotising soft tissue infections (NSTIs) are rare but life-threatening conditions associated with high mortality rates. This multicentre study aimed to identify admission variables associated with in-hospital mortality.

METHODS: This retrospective study included adult patients with surgically confirmed NSTIs treated at four high-volume academic referral centres in Italy between 2010 and 2024. Demographic, clinical, physiological, and laboratory variables available at hospital admission were analysed. Categorical variables were compared between survivors and non-survivors using the chi-square test or Fisher's exact test, while quantitative variables were compared using the Student's t test or Mann-Whitney U test. Multivariable logistic regression analyses were performed to identify independent predictors of in-hospital mortality. Results were reported as ORs with 95%CI. A

prognostic nomogram was developed from the final multivariable model, including variables independently associated with mortality.

RESULTS: A total of 379 patients were included. In-hospital mortality was 16.7%. In subgroup comparisons, mortality was 17.0% in necrotising fasciitis and 22.4% in Fournier's gangrene ($P = 0.275$). In the overall NSTI population, age (aOR 1.061, 95%CI 1.037-1.087) and serum creatinine at admission (aOR 1.301, 95%CI 1.040-1.629) were independently associated with mortality [...]

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Influence of frailty on short-term mortality in older patients with multiple trauma in the emergency department.

Çelik ■, Toker A, Biçer S, Cebecio ■/u ■K

PURPOSE: To investigate the association between frailty and 28-day mortality in older patients presenting to the emergency department with multiple trauma and to assess the discriminative performance of PRISMA-7 and the Trauma-Specific Frailty Index (TSFI) in frailty assessment in relation to commonly used trauma severity scores.

METHODS: This prospective observational study included patients aged ≥ 65 years with multiple trauma admitted to the emergency department between July 1 and August 31, 2024. Frailty was assessed using PRISMA-7 and TSFI. Patients were stratified into frail and non-frail groups and further categorized according to age groups (65-74, 75-84, and ≥ 85 years). Clinical characteristics, trauma severity scores, and outcomes, including 28-day mortality, were compared. Receiver operating characteristic (ROC) analysis was used to evaluate discriminative performance, and logistic regression analysis was performed to assess the association between scores and mortality.

RESULTS: A total of 144 patients were included. Frailty was significantly associated with increased 28-day mortality, higher ICU admission rates, and longer hospital stays ($p < 0.001$). Mortality increased with age, reaching 30.0% in patients aged ≥ 85 years ($p = 0.006$). Frailty scores (PRISMA-7 and TSFI) increased significantly with age, whereas ISS did not differ significantly across age groups. R [...]

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Mortality prediction in older adults with earthquake-related trauma: a model combining MEWS and laboratory biomarkers.

Ozen A, Acehan S, Satar S, Bulut EA, Gulen M, N ■g ■z K, Sevd ■mbas S, Sonmez GO et al.

PURPOSE: This study aimed to identify laboratory parameters and clinical scores that can predict in-hospital mortality among patients aged 65 years and older who presented to the emergency department (ED) with earthquake-related trauma following the February 6 and 20, 2023 earthquakes.

METHODS: In this retrospective observational study, 362 patients with earthquake-related trauma between February 6-21, 2023, were included. Demographic characteristics, laboratory results, and clinical scores such as the Modified Early Warning Score (MEWS), Injury Severity Score (ISS), and Shock Index (SI) were analyzed.

RESULTS: Of the included patients, 53.9% ($n = 195$) were female, and the median age was 73 years (IQR: 69-80). The overall mortality rate was 9.4%. Multivariable logistic regression analysis identified MEWS (OR: 3.535; 95% CI: 2.397-5.212; $p < 0.001$), urea (OR: 1.009; 95% CI: 1.002-1.015; $p = 0.008$), and hs-Troponin I (OR: 1.001; 95% CI: 1.000-1.002; $p = 0.016$) as independent predictors of mortality. Receiver operating characteristic (ROC) analysis demonstrated an AUC of 0.936 for the combined model including MEWS, urea, and hs-Troponin I (95% CI: 0.875-0.996; $p < 0.001$). Among the variables, MEWS showed the highest AUC (0.903; 95% CI: 0.829-0.976). The optimal cutoff value for MEWS was determined as 1.5, with a sensitivity of 70.6% and a specificity of 97.6%.

CONCLUSIONS: In o [...]

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Impact of Preoperative Osteoporotic Medications on Total Hip Arthroplasty Outcomes.

Antonioli SS, Ruff G, Leung N, Patel A, Schwarzkopf R, Cohen-Rosenblum A

BACKGROUND: Osteoporosis (OP) is a common comorbidity in patients undergoing total hip arthroplasty (THA) and is a known risk factor for poor postoperative outcomes such as periprosthetic fracture (PPF). The impact of preoperative OP medications in patients with OP undergoing THA remains unclear. This study aimed to compare THA outcomes by OP diagnosis and preoperative bone strengthening medication usage.

METHODS: This was a retrospective review of primary elective THAs from June 2011-January 2024. Patients were stratified by OP diagnosis and OP medication usage, then propensity matched in a 1:2:3 ratio by age, sex, body mass index, and comorbidities. The resulting cohorts: (1) OP and medication usage for at least 1 year preoperatively and within 7 years of the procedure ($n = 296$), (2) OP and no medication usage ($n = 592$), and (3) no diagnosis of OP and no medication usage ($n = 888$) were then compared for postoperative outcomes.

RESULTS: Patients in cohort 1 were more likely to visit the emergency department ($P = .043$) or be readmitted for a surgical complication ($P = .007$) within 90 days. They were more likely to undergo posterior approach than cohorts 2 and 3 ($P = .016$), had the highest occurrence of cemented femoral stems (38.2%, $P < .001$), and had a higher incidence of revision for PPF (3.4%, vs 0.8% [cohort 2] and 1.2% [cohort 3], $P = .009$). Of the 12 revisions due to pe [...]

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